

SHOP – PRODUCT SALE & BILLING

Input Prompts for Project Design

BCA 1st-Year Assignment

Beginner-Friendly Project

1 ■■■ OVERALL PROJECT

Copy this prompt:

Create a simple and beginner-friendly software system for Shop – Product Sale & Billing based on the following features: Customer Details, Product Details, Quantity, Sale, and Bill Calculation. The system should allow the shopkeeper to enter customer information, select products, enter quantity, calculate the total amount, and generate a bill. Explain the system in simple language suitable for a BCA 1st-year student. Provide the project objective, requirements, flowchart, algorithm, and ER diagram separately. Keep the design simple and easy to understand.

2 ■■■ SYSTEM REQUIREMENTS

Copy this prompt:

Prepare the System Requirements for a project titled “Shop – Product Sale & Billing” using these features: Customer Details, Product Details, Quantity, Sale, and Bill Calculation. Divide the requirements into Functional Requirements and Non-Functional Requirements. Include requirements such as entering customer details, storing product details, entering quantity, recording sales, calculating subtotal and total bill, and generating a bill. Write everything in simple and clear language suitable for a BCA 1st-year assignment.

Reference / Expected Output:

Functional Requirements

- Add and store customer details.
- Add and store product details.
- Enter the quantity of products purchased.
- Record product sales.
- Calculate the amount for each product.
- Calculate the total bill.
- Display/generate the final bill.

Non-Functional Requirements

- Easy to use.
- Fast calculation.
- Accurate billing.
- Secure storage of data.
- Simple and user-friendly interface.

3 ■■■ FLOWCHART

Copy this prompt:

Create a simple and properly structured flowchart for the project “Shop – Product Sale & Billing” based on these features: Customer Details, Product Details, Quantity, Sale, and Bill Calculation. The flowchart should start with Start, then enter customer details, enter/select product details, enter quantity, calculate the product amount using $\text{Quantity} \times \text{Product Price}$, record the sale, calculate the total bill, display the bill, and end with Stop. Use standard flowchart symbols such as oval for Start/Stop, parallelogram for input/output, rectangle for processing, and diamond for decisions. Make the flowchart suitable for a BCA 1st-year assignment and keep it simple.

Reference / Expected Output:

```
Basic Flow
Start
↓
Enter Customer Details
↓
Enter/Select Product Details
↓
Enter Quantity
↓
Calculate Amount
Amount = Price × Quantity
↓
Record Sale
↓
Calculate Total Bill
↓
Display Bill
↓
Stop

Optional decision:
“Add another product?”
→ Yes → Select another product
→ No → Calculate total bill
```

4 ■ ■ ■ ALGORITHM

Copy this prompt:

Write a simple step-by-step algorithm for “Shop – Product Sale & Billing” using the following features: Customer Details, Product Details, Quantity, Sale, and Bill Calculation. The algorithm should include entering customer details, selecting/entering product details, entering quantity, calculating the product amount using $\text{Price} \times \text{Quantity}$, recording the sale, calculating the total bill, displaying the final bill, and stopping the process. Use numbered steps and simple language suitable for a BCA 1st-year student.

Reference / Expected Output:

Expected Algorithm

Algorithm: Shop Product Sale & Billing

1. Start.
2. Enter customer details.
3. Enter or select product details.
4. Enter the quantity purchased.
5. Get the price of the product.
6. Calculate product amount:
 $\text{Amount} = \text{Price} \times \text{Quantity}$
7. Record the sale details.
8. Check whether the customer wants to purchase another product.
9. If yes, repeat the product and quantity entry.
10. If no, calculate the total bill.
11. Display the final bill.
12. Stop.

5 ■■ ER DIAGRAM

Copy this prompt:

Create a simple and correct ER (Entity Relationship) Diagram for a Shop – Product Sale & Billing System based on these features: Customer Details, Product Details, Quantity, Sale, and Bill Calculation. Use the entities Customer, Product, Sale, and Bill. Include suitable attributes such as Customer_ID, Customer_Name, Phone, Product_ID, Product_Name, Price, Sale_ID, Sale_Date, Quantity, Amount, Bill_ID, Total_Amount, etc. Show primary keys clearly and show the relationships between the entities using proper cardinality such as one-to-many. Keep the ER diagram simple, neat, and suitable for a BCA 1st-year assignment. Do not add unnecessary entities or attributes.

Reference / Expected Output:

Suggested ER Structure

CUSTOMER

- Customer_ID (PK)
- Customer_Name
- Phone

makes

↓

SALE

- Sale_ID (PK)
- Sale_Date
- Customer_ID (FK)
- Product_ID (FK)
- Quantity
- Amount

contains

↓

PRODUCT

- Product_ID (PK)
- Product_Name
- Price

SALE

■

■ generates

↓

BILL

- Bill_ID (PK)
- Sale_ID (FK)
- Total_Amount

Note: Show the relationships using proper ER notation and cardinality (such as 1:M) in the final diagram.

Tip: Use these prompts one at a time in an AI/design tool to generate the corresponding project output. The prompts are intentionally kept simple for a BCA 1st-year assignment.